surveillance and long-range strikes. These UAVs operate semi-autonomously, coordinated through the combat cloud and AI mission management algorithms.

The FCAS program is more than a technological initiative; it is a strategic collaboration among European nations to achieve aerospace sovereignty, reduce dependency on non-European defense systems, and pave the way for future sixth-generation air combat systems. The combination of AI-enabled autonomy, stealth design, and multi-domain integration makes FCAS a model for the next phase of UAV and manned–unmanned cooperative warfare.

Figure 5.7 illustrates the Future Combat Air System (FCAS), emphasizing its System of Systems (SoS) architecture that integrates a Next Generation Fighter (NGF), multiple Remote Carrier Drones (RCs), and a centralized Combat Cloud Network. The NGF acts as the command hub, while the RCs perform diverse roles such as surveillance, electronic warfare, and precision strikes, all interconnected through the Combat Cloud for real-time data sharing. This configuration demonstrates FCAS's core principle, AI-driven, multi-platform collaboration, where manned and unmanned assets function cohesively to achieve information dominance and adaptive mission execution across air, space, and cyber domains. The figure effectively portrays Europe's vision of integrated, autonomous airpower, setting a technological benchmark for future sixth-generation air combat.